



Inclusive design

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EDITORIAL

Inclusive design

The 1990s saw a sea change in thinking about assistive technology and design for disability in response to a growing awareness of demographic ageing as a global trend, and the widespread desire among the disabled community for inclusion within the mainstream of consumer society. Although initially driven by relatively small groups of people in the US, Europe and Japan responding to local conditions, this rapidly became international in character and influence. Although a range of terms was used – universal design, inclusive design, design for all, gerontechnology, transgenerational design – the key and common shift in thinking was to replace the view that people are disabled by physical and mental impediments with the more radical proposal that people are disabled by designs and environments that do not take account of the full range of human capabilities. As a result, links were formed between user organisations, government and the design community leading to new products, services and legislation reflecting this shift in thinking.

A practical lead was given by Sweden, where innovative designers such as Maria Benktzon of Ergonomi Design Gruppen collaborated with the Swedish Health Service to develop high quality assistive products based on extensive user research and contemporary design values. This development was triggered by a 1968 seminar at the Design School in Stockholm involving Victor Papanek and Andrzej Pawlowski. Papanek also spoke at a similar design seminar at the Royal College of Art, London, in 1976, which explored the subject of Design for Need, and between 1979 and 1982, American designer Patricia Moore of Raymond Leowy's New York office conducted an influential experiment where she transformed herself into an elderly woman to reveal the inadequacy of much design.

More recently, advances have been made in Japan, the US and Europe that are embedding this thinking in industry and education, including the introduction of BS 7000-6 on inclusive design management, the transformation of BT from a company with a special section devoted to 'older and disabled consumers' to a company with a core inclusive design strategy, and the launch of a Knowledge Transfer programme in inclusive design under the UK's government-funded EQUAL programme.

The EQUAL programme, begun in the 1990s, demonstrated that by keeping researchers close to research users and beneficiaries, as well as ensuring that they were networked to policy makers, major improvements could be achieved very quickly both locally and nationally. It succeeded in encouraging a new generation of researchers, both from engineering and industrial design backgrounds, to participate in an already active enthusiastic research community. Two new biennial international conference series were also established to support the programme: INCLUDE, with its inaugural event in London in 2001 and a sixth meeting planned for 2011; and

CWUAAT held in Cambridge and meeting in the alternate years, with its fifth workshop being held in 2010.

The worldwide Inclusive Design community has also been supported over recent years by an increasing number of events; including those organised by the International Association for Universal Design, who are running their third international conference in 2010 in Hamamatsu, Japan; and the Institute for Human Centered Design, with their Designing for the 21st Century conference in Rio de Janeiro. As this community of research and practice expands it is timely to showcase the best research, as rigorously judged by reviewers primarily from that community. In response to a call for papers early in 2008, we received 56 expressions of interest resulting in 50 abstracts for papers. Of the 27 abstracts accepted, 19 full papers emerged for review and 14 were subsequently accepted for publication. These have been grouped to reflect active research topics in the field of Inclusive Design, namely: technology adoption, technology solutions, design education and design tools.

Technology adoption: Morris *et al.* explore the emerging differences between the elders of today and the even-larger population of tomorrow's elders, and their likely behaviours as wireless users; giving guidance for wireless technology providers. Macdonald discusses customer-centred approaches in Japan that have begun to address the needs of previously excluded or ill-considered market sectors through a field survey in Japan of recent product innovation in mobile phones in the older person's, children's, and mainstream markets from 2003 to 2007. Kose reports on progress in the development of housing design concepts acceptable to older users in Japan, and discusses the remaining issues to be resolved to bring about an inclusive society. Imai *et al.* report on an observational study of how older adults learn a new user interface in their daily lives and presents factors, of critical importance to designers, that may make their learning process easier. Lim proposes a new approach for designing technological products which considers age-related decline in ability along with both the formative age when the use of technologies is first learnt, and the 'technology era' in which this occurred.

Technology solutions: Udo and Fels demonstrate how the processes and practices associated with closed captioning and audio description, in their current form, violate some of the main principles of inclusive design and introduce an alternative process and set of practices. Mountain *et al.* explore the usability of a prototype technology to enable stroke survivors to independently undertake rehabilitation exercises at home, emphasising the importance of the interplay between technology development, clinical utility and usability when delivering new devices into practice.

Design education: Dong discusses approaches to the teaching of inclusive design at degree level, citing three pilot studies focused on effective user research methods in large class teaching, involving end-users in the design process, and developing case studies based on successful student projects.

Design tools: Marshall *et al.* address two common needs for designers working towards developing inclusive products and environments, namely data on users that is accessible, valid and applicable, and a means of utilising the data to assess the accessibility of designs during the early stages of development. Johnson *et al.* review existing data sets and identify the challenges inherent in measuring capability in a manner that is informative for work in Inclusive Design. The need for a design-relevant capability data set is identified along with a number of practical issues pertinent to measuring such data. Carse *et al.* investigate designers' use of biomechanical data to inform the process of Inclusive Design in the consumer packaging industry; identifying guidance for the development of future data. Gyi *et al.* report on a study of older users' understanding of computer-based models used at an early stage of product design; showing that they were able to identify the basic purpose or function of a familiar product from 2D line drawings. Afacan and Demirkan explore the potential of a priority-based approach for satisfying diverse users' needs, capabilities and expectations in a design process, targeted towards architectural design applications. Goodman-Deane *et al.* report on a study that examined aspects of design

practice; articulating the need for designers to educate clients in the value of inclusive design and for informal, cheap, yet insightful inclusive design methods.

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